

RCDEPS Colony Ship — Preliminary Design Review (PDR) v1.0

Purpose-built interstellar–interplanetary vehicle integrating Recursive Coherence-Driven Electrodynamic Propulsion with analog solar-thermal/steam power and closed-loop life support

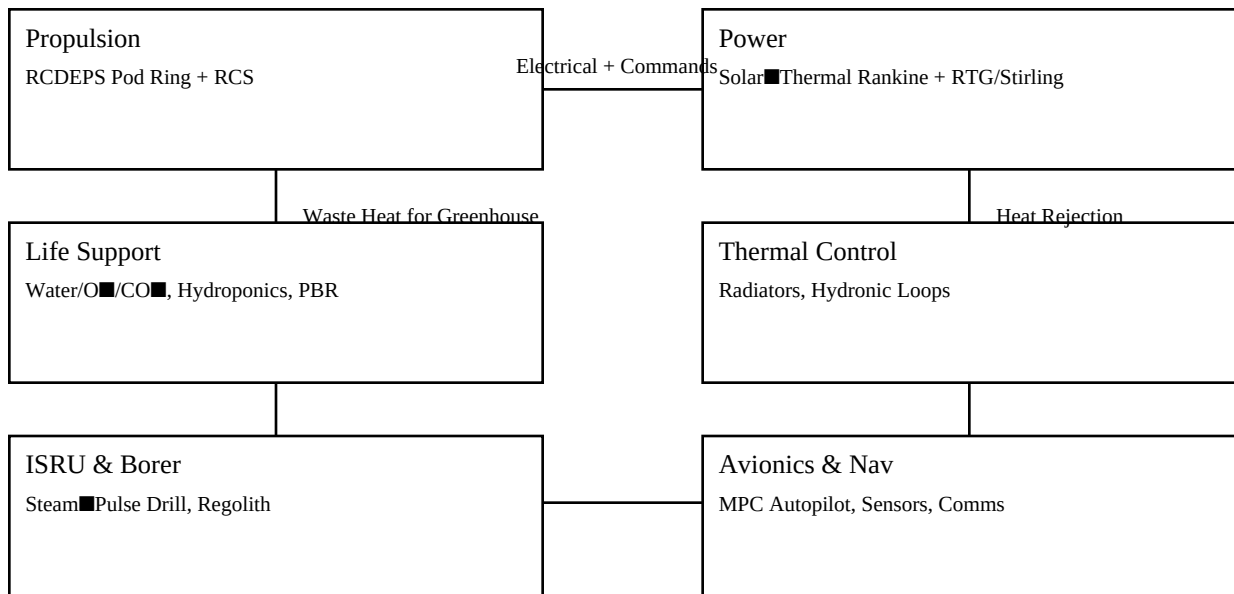
1. Mission Intent & Concept of Operations (ConOps)

The RCDEPS Colony Ship is designed to: (i) navigate complex obstacle fields and narrow “straits” with precise micro■thrust vectoring, (ii) brake and station-keep safely, (iii) operate on robust analog power using solar■thermal steam with passive fail■safes, (iv) sustain crew with closed■loop water, oxygen, and food via hydroponics/photobioreactors, (v) conduct in■situ resource utilization (ISRU), including steam■driven boring for subsurface habitats, and (vi) support long■duration colonization with modular construction toolchains and radiation■safe shelters.

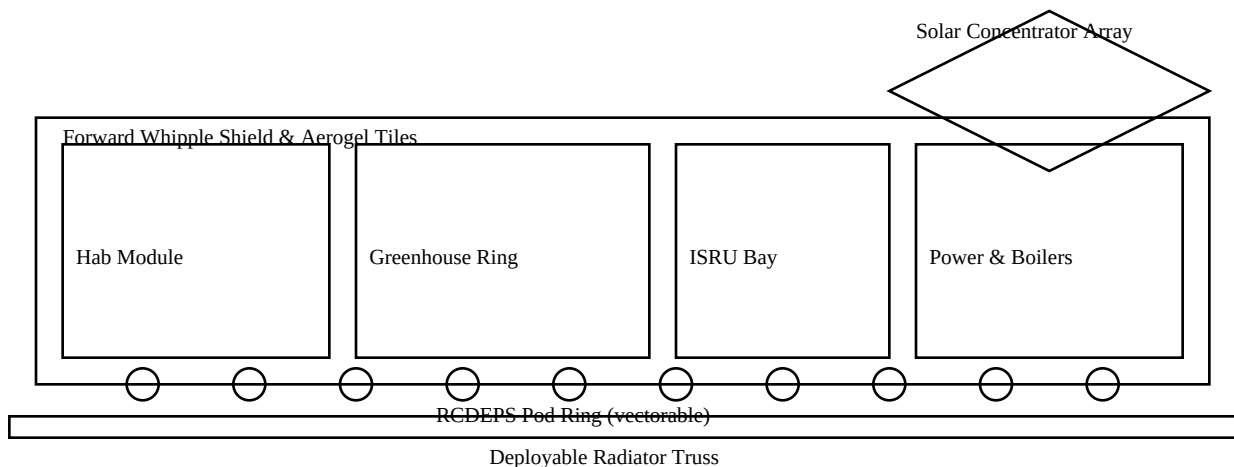
- Cruise: RCDEPS low■thrust vectoring; solar■thermal Rankine cycle provides power/heat; life support in steady state.
- Obstacle/“straits” mode: Lidar/radar mapping → MPC corridor flight at 10–20 Hz planning; thrust phase/delay modulation for tight control.
- Braking: RCDEPS reverse phasing; deployable photon sail drogue near stars; optional mag■sail for solar■wind drag; aerobrake where applicable.
- Colony ops: Steam borer creates vertical shafts and horizontal drifts; regolith sintering for bricks; greenhouses expand; water■shielded safe rooms built first.
- Safe mode: Defocus concentrators; close steam stop valve; dump pressure to thermal store; hold attitude with wheels; RTG/Stirling keep■alive.

2. System Architecture

2.1 Block Diagram



2.2 Ship Layout (Notional)



3. Top-Level Requirements

R■1 Propulsion: Vectorable micro■thrust with sign■reversing control; safe reverse for braking; independent attitude control.

R■2 Power: $\geq 100 \text{ kW}_e$ at 1 AU from solar■thermal; black■start and keep■alive from 1–5 kW RTG/Stirling; fully analog trip path.

R■3 Life Support: $\geq 95\text{--}99\%$ water recovery; O■ generation via PEM electrolysis + Sabatier loop; ≥ 30 days consumables buffer.

R■4 Agriculture: $\geq 10\text{--}20\text{ m}^2/\text{person}$ hydroponics (partial calories) + PBR for O■ smoothing; thermal/light integration.

R■5 ISRU: Steam■pulse borer $\geq 0.5\text{--}2.0\text{ m}^3/\text{h}$ in basaltic regolith; cuttings ejection and capture; liner installation.

R■6 Thermal: Radiators sized for full reject at 1 AU operation; hydronic cabin/greenhouse loops; phase■change sinks.

R■7 Safety: Whipple shielding; water■shielded safe room ($\geq 20\text{ cm}$); dual watchdogs; manual overrides; fire/HV interlocks.

R■8 Autonomy: MPC with $10\text{--}20\text{ Hz}$ update; multi■sensor fusion (star tracker, IMU, lidar/radar); failsafe hold.

4. Preliminary Sizing & Budgets

Assuming 500 m^2 concentrators at 1 AU: Thermal input $\approx 544\text{ kW}_{\text{th}}$, Electric $\approx 109\text{ kW}_{\text{e}}$ ($\eta_{\text{opt}}=0.8$, $\eta_{\text{elec}}=0.2$).

Radiator area at $T=350\text{ K}$, $\varepsilon=0.85$: $A_{\text{rad}} \approx 602\text{ m}^2$ to reject waste heat (conservative; some heat reused for habitat/greenhouse).

Life Support First-Cut (scalable):

Crew	Hydroponics Area (m^2 , partial diet)	Water Buffer (days @ 15 L/p/d)	Safe-room Water Shield (kg)
4 crew	12	30 d (60 L/d total)	2000
8 crew	20	30 d (120 L/d total)	2000
20 crew	50	30 d (300 L/d total)	2000

Propulsion envelope uses multiple RCDEPS pods (vectorable) for precise micro■thrust and torque. For cruise braking and stellar approach, deploy a photon sail drogue; for solar■wind interaction, a superconducting mag■sail loop provides additional drag. Attitude control via reaction wheels + CMGs; cold■gas microthrusters as fail■safe. Reverse thrust achieved by RCDEPS phase flips.

5. Subsystem Specifications

5.1 Propulsion & GNC

- RCDEPS Pods: $6\text{--}12$ around hull; $\pm 15^\circ$ vector range; closed■loop on amplitude β , phase ϕ , delay τ .
- Sensors: dual star trackers, rad■hard IMUs, CW lidar ($\leq 30\text{ km}$), Ka■band radar, dust/charge sensors, dosimeters.
- Control: Model■Predictive Control (MPC) $10\text{--}20\text{ Hz}$; inner attitude $20\text{--}50\text{ Hz}$; dual watchdogs (digital + analog).
- Obstacle Navigation: Occupancy grid $\rightarrow$ corridor generation $\rightarrow$ minimum■jerk trajectories; keep■out volumes.
- Braking: RCDEPS phase reversal; photon sail drogue near stars; optional mag■sail in solar wind.

5.2 Power (Analog Solar■Thermal Steam)

- Concentrators $400\text{--}800\text{ m}^2$; 2■axis tracking; safe defocus.
- Receiver/Boiler: molten■salt primary $\rightarrow$ water/steam secondary; mechanical overspeed trips; relief valves; hand■wheel overrides.
- Turbine/Alternator: 2■stage impulse; mechanical governor; flywheel; diode crowbar protection on DC link.
- Condenser/Radiators: $Q=\varepsilon\sigma A(T_{\text{■}}-T_{\text{space■}})$; hydronic loops for cabin/greenhouses; phase■change buffers.
- Storage/Black■Start: molten■salt $8\text{--}12\text{ h}$; $1\text{--}5\text{ kW}$ RTG/Stirling for keep■alive and black start.

5.3 ECLSS (Water–Air–Food)

- Water: vapor■compression distillation + multi■filtration + UV/AOP → ≥99% recovery; ≥30■day buffer.
- Oxygen/CO■: PEM electrolysis; Sabatier; solid amine beds; algae PBR smoothing.
- Hydroponics: vertical NFT + deep■water/aeroponics; daylight light■pipes + LED trims; nutrient reclaim.
- Food: partial calories fresh; staples from stores; expand racks post■landing.

5.4 ISRU & Steam■Pulse Borer

- Drill: percussive + rotary; steam■pulse ejector; adjustable standoff jets; liners installed in■line.
- Steam: 20–80 bar; cam■timed pulses; cyclones/baghouses; cuttings hoppers; sintering optics for bricks.
- Throughput: 0.5–2.0 m³/h (material dependent).

5.5 Thermal Control & Shielding

- Radiators: louvered; dust■resistant; variable emissivity options.
- Shielding: Whipple + aerogel; safe room: ≥20 cm water; regolith berms post■landing.

6. Interface Control (ICD) — Excerpts

Interface	From → To	Medium	Notes
Power bus	Turbine Gen → DC/DC bus	3■phase AC / DC	Analog governor; crowbar trip
Thermal loop	Boiler → Condenser → Radiators	HT salt / Water	Isolation valves; temp sensors
Propulsion cmd	Avionics → RCDEPS pods	Fiber + copper	β, φ, τ setpoints; hard inhibit line
Sensor fusion	Star/IMU/Lidar → Avionics	Fiber/CAN	Time■syncd; PPS timebase
ISRU steam	Boiler → Drill head	Steam line	Pulse valves; ejector return

7. Risk Register (Top 8)

- 1) RCDEPS thrust uncertainty → photon calibrator; null runs; independent replication.
- 2) HV arcs/corona → rounded electrodes; interlocks; RF shielding; vacuum ops.
- 3) Thermal margins → radiator oversize; PCMs; variable■emittance surfaces.
- 4) Dust impacts → Whipple/aerogel; charged■wire guard; replaceable tiles.
- 5) Greenhouse yield variance → mixed crops; LED redundancy; diagnostics.
- 6) Borer clogging → purge cycles; jet standoffs; maintenance access.
- 7) Attitude saturation → wheel desats; microthrusters; momentum management.
- 8) Single■points → dual watchdogs; manual overrides; spares; cold■spares.

8. Verification & Validation (V&V;) Plan

- 5 kW_e solar■thermal bench 500■hour endurance with trips/black■start drills.
- RCDEPS vacuum thrust campaign with artefact controls; phase■flip; orientation/B■field sweeps.
- 120■day closed■loop ECLSS/hydroponics demo with crew analogs; 99% water recovery.
- Steam borer basalt test; throughput, cuttings handling; liner install.
- Integrated demo: concentrator → steam → electric → RCDEPS pod + greenhouse rack on shared bus.

9. Bill of Materials (Summary)

Category	Key Items
Propulsion	RCDEPS pods (x12), vector mounts, reaction wheels, CMGs, microthrusters
Power	Segmented mirrors, receiver/boiler, molten■salt tanks, turbine/alt, condensers, radiators

ECLSS	PEM electrolyzers, Sabatier reactor, water processors, PBRs, hydroponic racks, LED rails
ISRU	Steam borer head, pulse valves, cyclones, baghouses, liners, sintering optics
Avionics	Rad■hard compute, star trackers, IMUs, lidar/radar, timebase, comms
Structures	Whipple shields, aerogel tiles, trusses, hull sections, safe■room tankage
Safety	Interlocks, E■stops, μ■metal, PPE, fire suppression

10. Next Actions

- A) Freeze PDR baselines; open long■lead procurement (mirrors, radiators, pumps/valves).
- B) Build 5 kW_e solar■thermal bench; commence RCDEPS vacuum campaign; start greenhouse pilot.
- C) Generate detailed ICDs, harness drawings, and safety analyses for CDR; define assembly flow and ATLO plan.

— End of PDR v1.0 —